

CO3_AT-1 Case Studies (1–5)

Case Study 1 – Employee Salaries

Given Data:

4, 5, 5, 6, 7, 8, 8, 9, 30

Q1. Mean, Median & Mode

Formula:

$$\text{Mean} = \Sigma x/n$$

$$\Sigma x = 4+5+5+6+7+8+8+9+30 = 82$$

$$n = 9$$

$$\text{Mean} = 82/9 = 9.11 \text{ lakhs}$$

Median:

Ordered data = 4,5,5,6,7,8,8,9,30

$$\text{Median position} = (9+1)/2 = 5\text{th value} = 7$$

Mode:

5 appears 2 times

8 appears 2 times

Mode = 5 and 8

Q2. Range

Formula:

$$\text{Range} = \text{Maximum} - \text{Minimum}$$

$$= 30 - 4$$

$$= 26 \text{ lakhs}$$

Q3. Variance & Standard Deviation

Formula:

$$\text{Variance} = \Sigma (x-\mu)^2/n$$

$$\text{SD} = \sqrt{\text{Variance}}$$

$$\text{Mean } \mu = 9.11$$

x	$x-\mu$	$(x-\mu)^2$
4	-5.11	26.12
5	-4.11	16.90
5	-4.11	16.90
6	-3.11	9.68
7	-2.11	4.46
8	-1.11	1.23
8	-1.11	1.23
9	-0.11	0.01
30	20.89	436.35

$$\Sigma(x-\mu)^2 = 512.88$$

$$\text{Variance} = 512.88/9 = 56.99$$

$$\text{SD} = \sqrt{56.99} = 7.55$$

Q4. Outlier

30 lakhs is an outlier because it is much larger than the remaining salaries.

Q5. Conclusion

Median (7 lakhs) is the best measure because the outlier inflates the mean.

Case Study 2 – Website Response Time

Given Data:

120,125,128,130,132,135,140,145,150,300

Q1. Mean & Median

$$\Sigma x = 120 + 125 + 128 + 130 + 132 + 135 + 140 + 145 + 150 + 300 = 1505$$

$$\text{Mean} = 1505 / 10 = 150.5 \text{ ms}$$

$$\text{Median} = (132 + 135) / 2 = 133.5 \text{ ms}$$

Q2. Range

$$\text{Range} = 300 - 120 = 180 \text{ ms}$$

Q3. Variance & SD

Formula:

$$\text{Variance} = \Sigma(x - \mu)^2 / n$$

$$\text{Mean} = 150.5$$

x	$x - \mu$	$(x - \mu)^2$
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120	-30.5	930.25
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125	-25.5	650.25
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128	-22.5	506.25
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130	-20.5	420.25
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132	-18.5	342.25
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135	-15.5	240.25
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140	-10.5	110.25
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145	-5.5	30.25
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150	-0.5	0.25
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300	149.5	22350.25
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$$\Sigma(x - \mu)^2 = 25580.50$$

$$\text{Variance} = 25580.50 / 10 = 2558.05$$

$$\text{SD} = \sqrt{2558.05} = 50.58 \text{ ms}$$

Q4. Skewness

The data is positively (right) skewed due to the value 300.

Q5. Conclusion

Median is preferred because it is unaffected by the outlier.

Case Study 3 – Student Performance

Given Data:

42,45,48,50,52,52,55,58,60,62,65,95

Q1. Quartiles

Lower half: 42,45,48,50,52,52

$$Q1 = (48 + 50) / 2 = 49$$

$$\text{Median}(Q2) = (52 + 55) / 2 = 53.5$$

Upper half: 55,58,60,62,65,95

$$Q3 = (60 + 62) / 2 = 61$$

Q2. IQR

$$\text{IQR} = Q3 - Q1$$

$$= 61 - 49$$

$$= 12$$

Q3. Outlier

$$\text{Lower Limit} = 49 - 1.5 \times 12$$

$$= 49 - 18$$

$$= 31$$

$$\text{Upper Limit} = 61 + 18$$

$$= 79$$

$$95 > 79$$

Therefore 95 is an outlier.

Q4. Mean & Median

$$\Sigma x = 684$$

$$\text{Mean} = 684 / 12 = 57$$

$$\text{Median} = 53.5$$

Q5. Conclusion

The outlier increases the mean but has very little effect on the median.

Case Study 4 – E-Commerce Orders

Given Data:

18,20,22,22,24,25,26,28,30,35

Q1. Mean, Median & Mode

$$\Sigma x = 250$$

$$\text{Mean} = 250/10 = 25$$

$$\text{Median} = (24 + 25)/2 = 24.5$$

$$\text{Mode} = 22$$

Q2. Variance & SD

$$\text{Mean} = 25$$

$$x \quad x - \mu \quad (x - \mu)^2$$

$$18 \quad -7 \quad 49$$

$$20 \quad -5 \quad 25$$

$$22 \quad -3 \quad 9$$

$$22 \quad -3 \quad 9$$

$$24 \quad -1 \quad 1$$

$$25 \quad 0 \quad 0$$

$$26 \quad 1 \quad 1$$

$$28 \quad 3 \quad 9$$

$$30 \quad 5 \quad 25$$

$$35 \quad 10 \quad 100$$

$$\Sigma(x - \mu)^2 = 228$$

$$\text{Variance} = 228/10 = 22.8$$

$$\text{SD} = \sqrt{22.8} = 4.77$$

Q3. Coefficient of Variation

$$\text{CV} = (\text{SD}/\text{Mean}) \times 100$$

$$=(4.77/25)\times 100$$

$$=19.08\%\approx 19.1\%$$

Q4. Interpretation

Orders deviate by about 4.77 from the average.

Q5. Conclusion

The order data shows moderate variability.

Case Study 5 – Comparing Two Data Science Models

Given Data:

Model A: 2, 3, 3, 4, 4, 5, 5, 6

Model B: 1, 1, 2, 4, 5, 7, 8, 9

Q1. Mean

Model A:

$$\Sigma x = 32$$

$$\text{Mean} = 32/8 = 4$$

Model B:

$$\Sigma x = 37$$

$$\text{Mean} = 37/8 = 4.625$$

Q2. Variance

Model A ($\mu = 4$)

$(x - \mu)^2$:

4, 1, 1, 0, 0, 1, 1, 4

$$\Sigma = 12$$

$$\text{Variance} = 12/8 = 1.5$$

Model B ($\mu = 4.625$)

$$\Sigma(x - \mu)^2 = 73.875$$

$$\text{Variance} = 73.875/8 = 9.23$$

Q3. Standard Deviation

Model A:

$$\text{SD} = \sqrt{1.5} = 1.22$$

Model B:

$$\text{SD} = \sqrt{9.23} = 3.04$$

Q4. Consistency

Model A is more consistent because it has the lower standard deviation.

Q5. Conclusion

A model with a lower mean error is not always better; consistency should also be considered.